

Random variable

↑

$$n = 26$$

↑

But $n \rightarrow \{H, T\}$

↑

↑ D.R.V.

Random variable

↑

C.R.V.

→

'class/room'

↑

162 | 168 - -- Real

Date → Product category ← D.R.V.
↑ ↓
[A, B, C]

Age ← C.R.V.
{ 0 ≤ n

Rolling a dice

$$n = \{1, 2, 3, 4, 5, 6\}$$

↑

$n = (1 \text{ to } 6)$

⊕ outliers → ?

Y = height of a student

↑
→ {122, 146.4, 132.5, ...}

↑
look good

Find

→ outliers

eg $T_x = \{50, 55, \dots, 2000\}$

↑
regular

eg Salary = {22, 25, 28, 125, 59}

↓ ↑ off
$$\text{Avg Sal} = \frac{22 + 25 + 28 + 125 + 59}{5}$$

↑
{12, 25}

↑
? outliers → 13110 → 13110
↓ many? → (i) Human error
⇒ 112
↑
Mean

$$\Rightarrow X \rightarrow$$

(eg 1)

$$X \rightarrow \begin{matrix} H & T \\ \downarrow & \downarrow \\ P(X=H) & 1/2 & 1/2 \end{matrix}$$

$$\sum P(X=x) = 1$$

$$X \rightarrow \text{P.O.R.V}$$

easy

(eg 2) $X \rightarrow$ Rolling a dice

$$X = \{1, 2, 3, 4, 5, 6\}$$

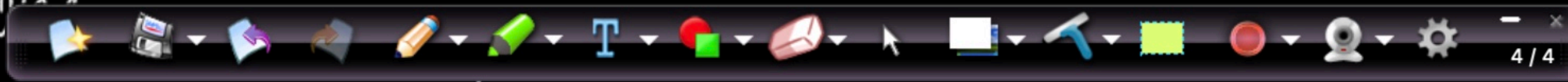
$$(i) \quad P(X < 2) = \frac{1}{6} \Rightarrow \left(\frac{1}{6}\right)$$

$$(ii) \quad P(X = \text{even}) = \frac{3}{6} = \frac{1}{2}$$

$$(iii) \quad P(X = \text{odd}) = \frac{3}{6} = \frac{1}{2}$$

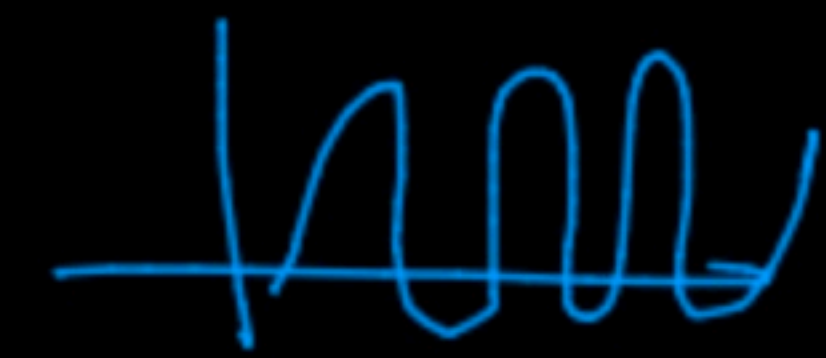
$$P = \frac{\text{favourable}}{\text{total event}}$$

CLASS ROOM



$\sin x$

+



$\xrightarrow{\text{C.R.V}}$

$$u = \{162, 165, \dots\}$$

Histogram

BPU

$f(x)$

Area

✓ $120 - 140 \rightarrow 23$

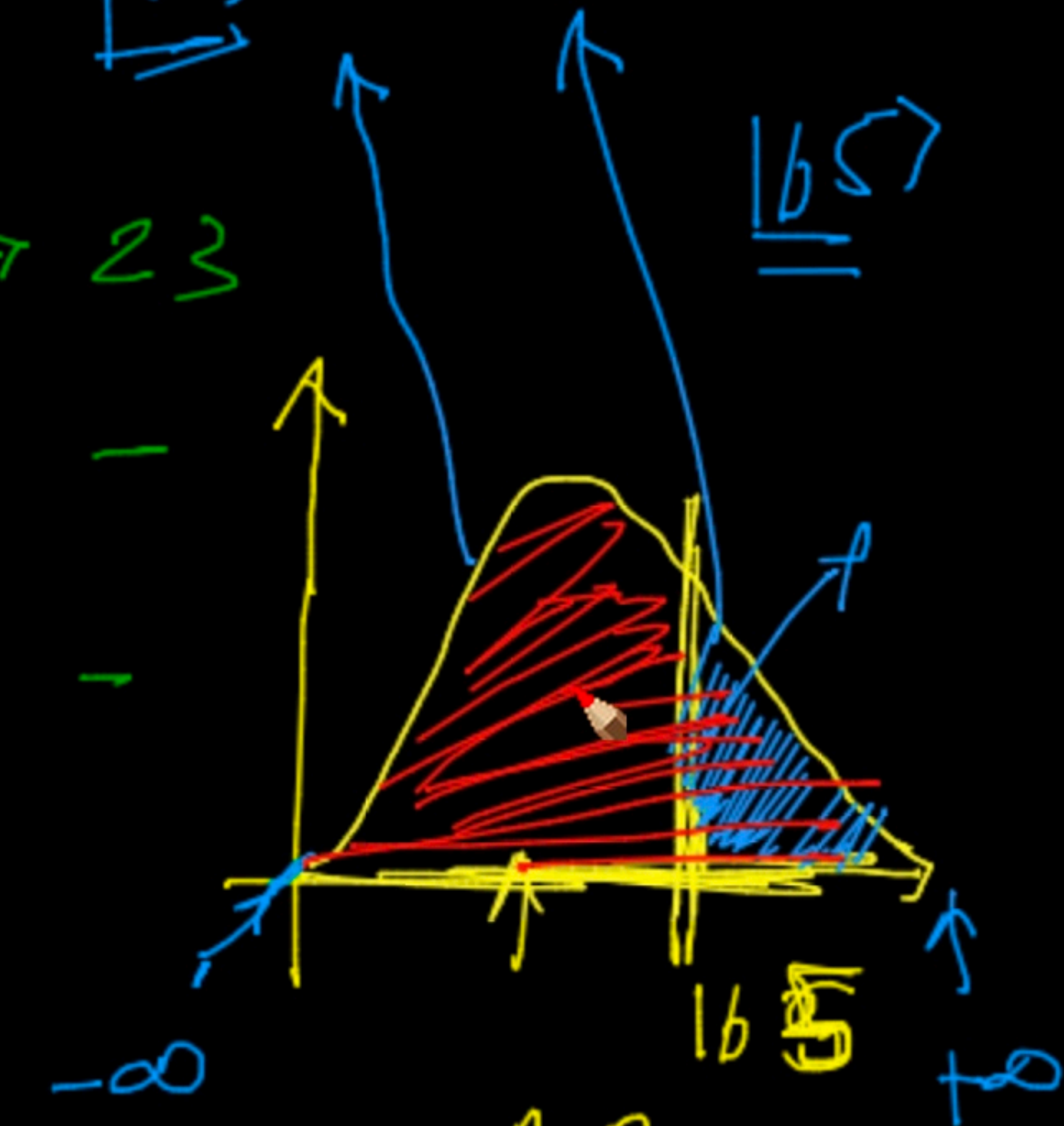
→ $140 - 150 - -$

$150 - 160 - -$

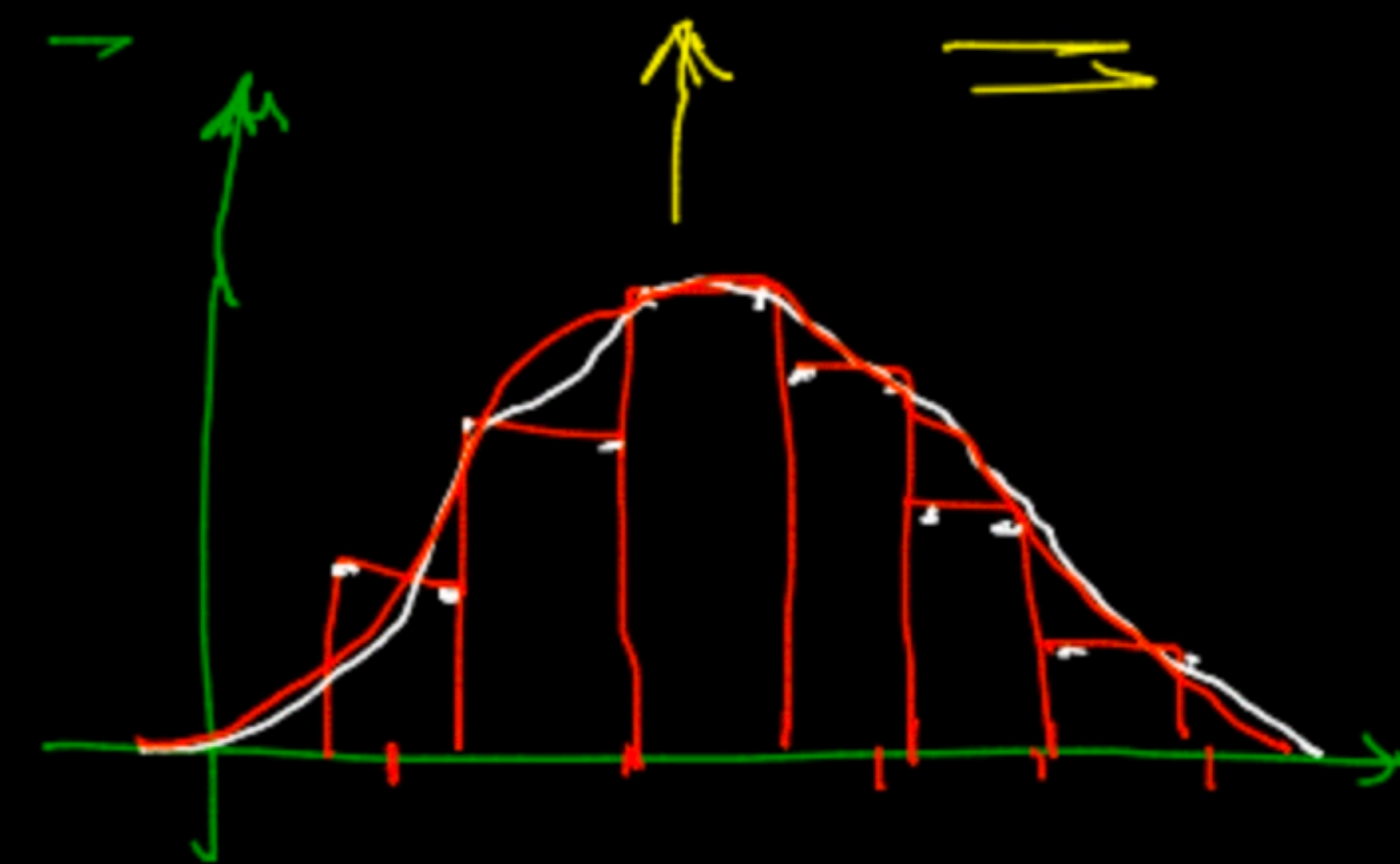
$160 - 170 -$

$170 - 180 -$

$> 180 -$



P.D.



→ $P(x > 165) = ? \rightarrow \text{easy}$

$P(146 < x < 155) = ? \rightarrow \text{easy}$

$\int_{-\infty}^{+\infty} f(x) dx = 1$

$$\int_{-\infty}^{\infty} f(x) dx$$

165. $\uparrow$

0.45

point $\rightarrow$ how to know $\rightarrow f(x)$

$\Downarrow$

well known distribution

$\rightarrow$ (i) Gaussian distribution / normal distribution

$\Downarrow$

$$X \sim N(\mu, \sigma^2)$$



plot

height

$$X \sim N(0, 1)$$

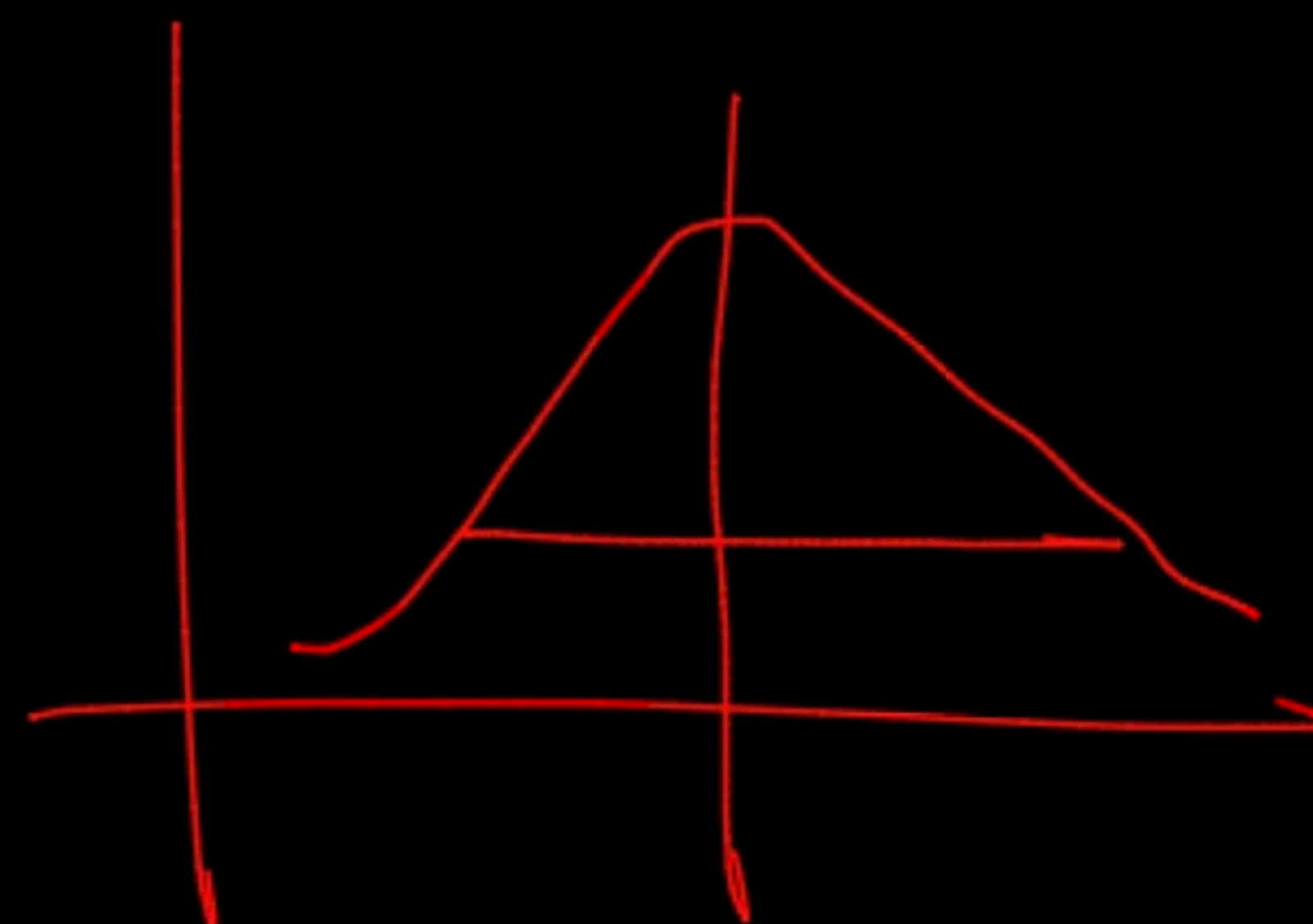
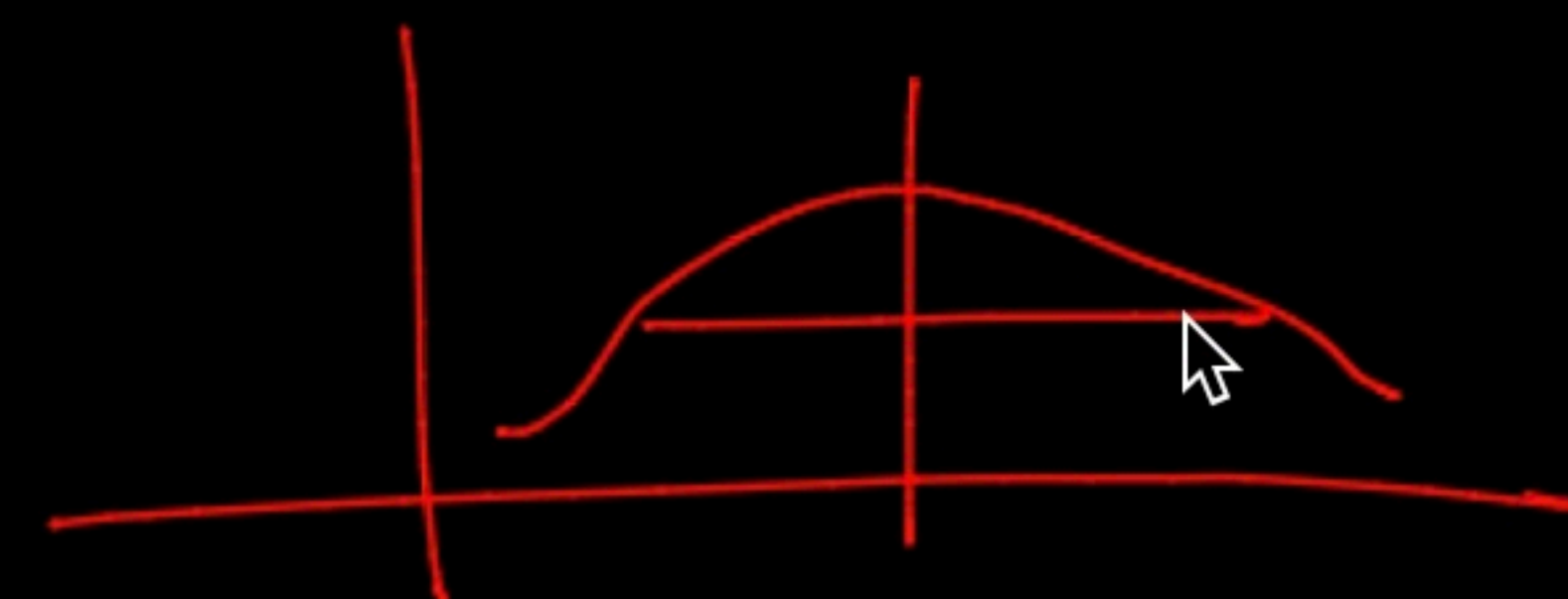
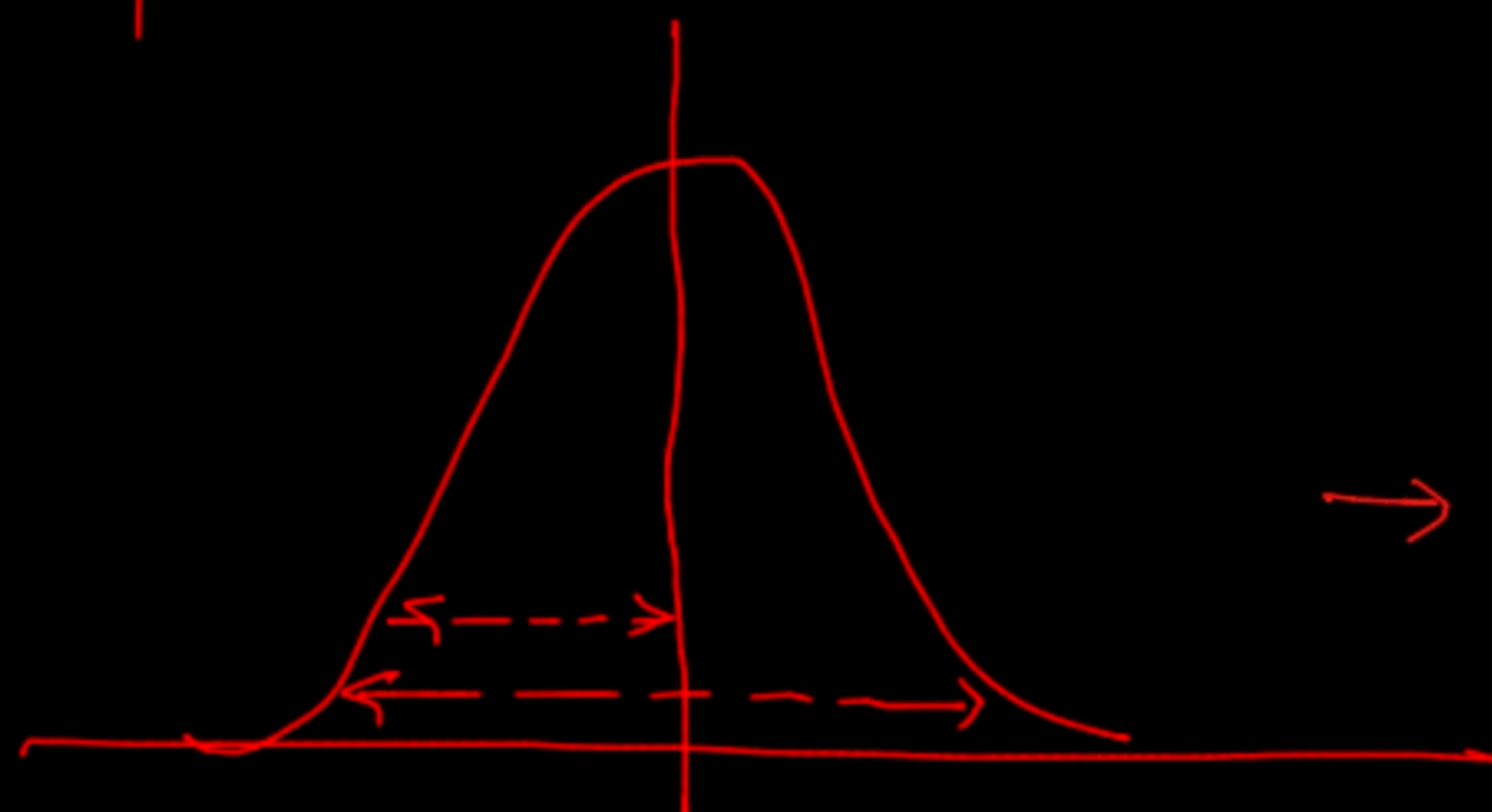
2

variance

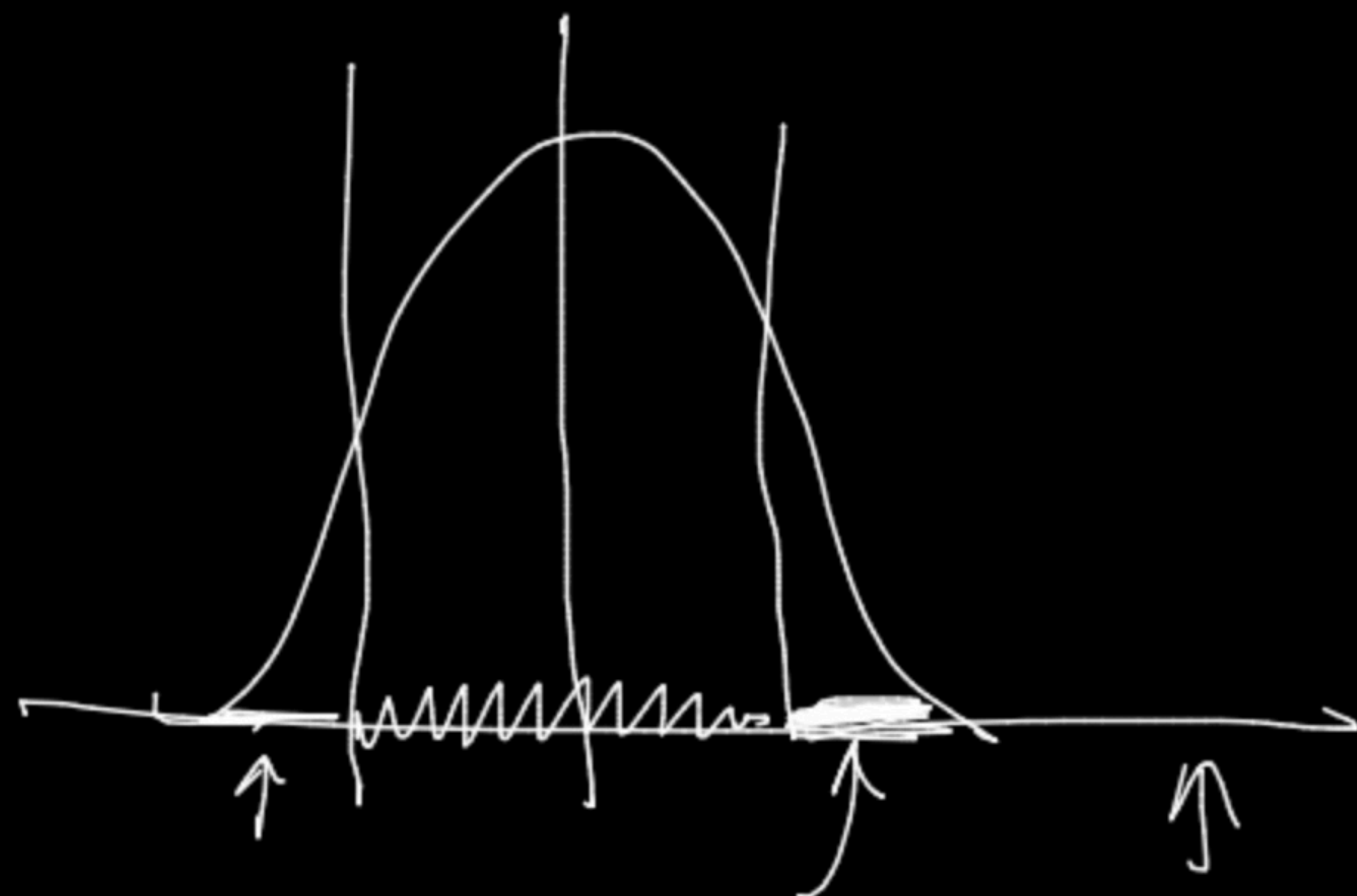


spread of data

mean

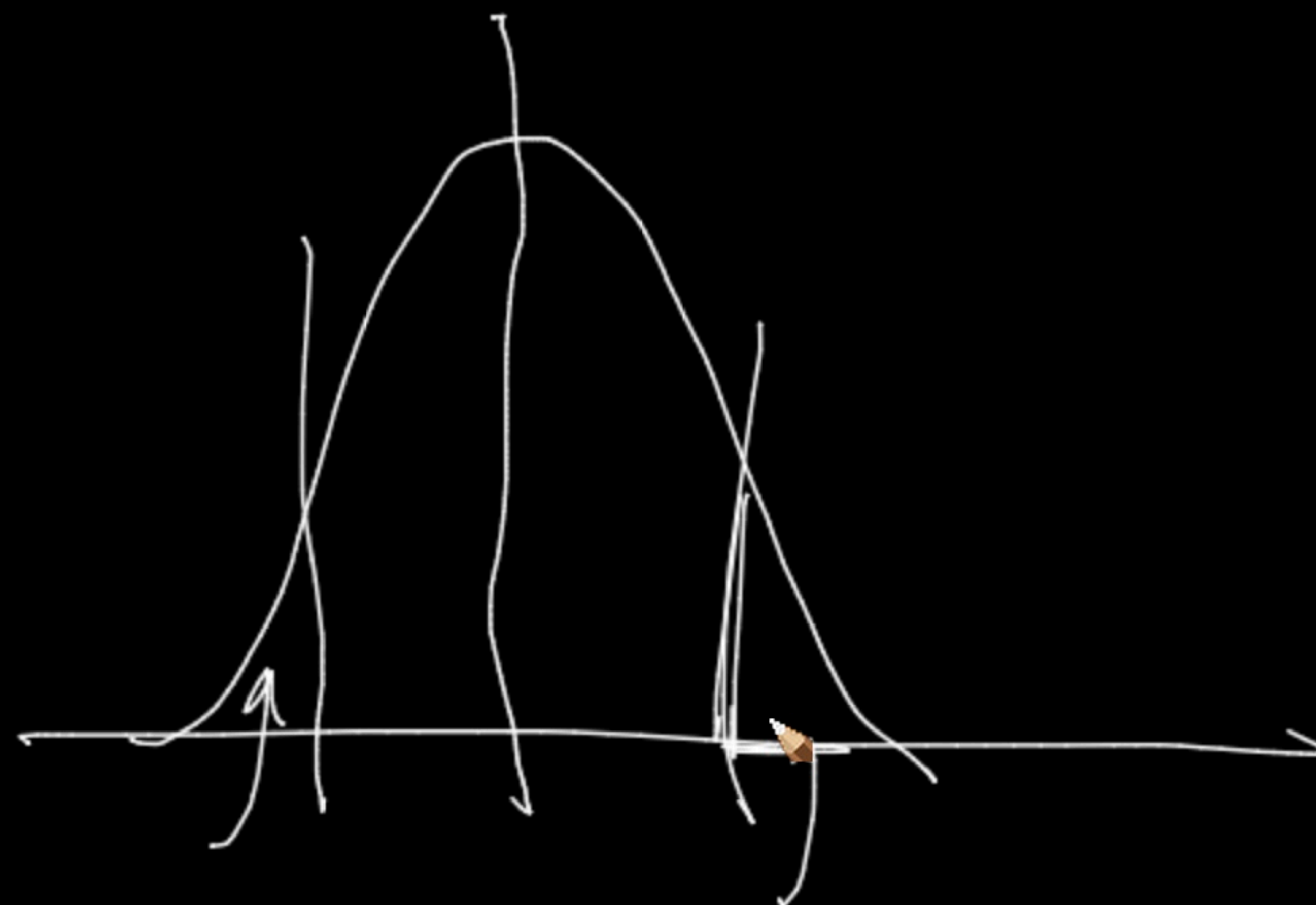


$\frac{d_i}{n}$



h_i

ρ_{income}



PDF

$$P(x=x) = P(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

het

$$\mu=0, \sigma^2=1, \sigma=1$$



$$P(x) = \frac{1}{\sqrt{2\pi}} * \exp \left\{ -\frac{(x)^2}{2} \right\}$$

plot

$$P(x=x) \approx \exp(-x^2) \Rightarrow$$

$$f = e^{-x^2}$$

Conclusion

- ↑ (1) x moves away from μ : γ reduces
- ✓ (2) Symmetric
- (3) Bell shape curve $\rightarrow e(-x^2)$
- (4) x moves close $\rightarrow$ mean $\rightarrow \underline{\gamma} \uparrow$

↓↓

Distribution $\rightarrow$ mean!

Case 1



XYZ
↑
{Slack}

Task
↑

order t-shirts → ?
=

S → 304
M → 5.15
L → 5.11

2 way → { 1) manually

→ collect

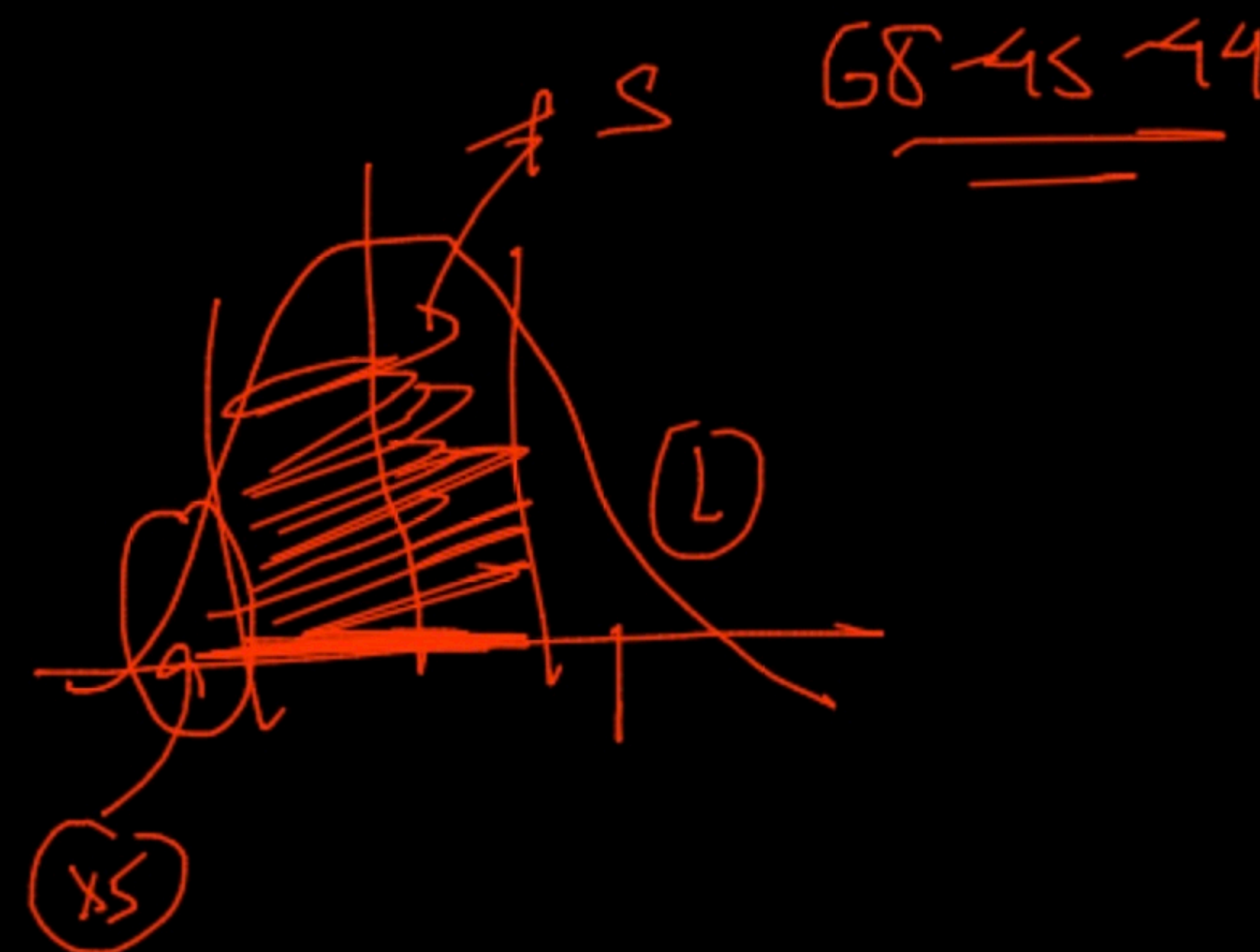
↓

h

↓

time ↑

task



$$P(X > 165)$$

$$\Rightarrow \int_{165}^{\infty} f(u) du$$

165

↑

↓↓

$$\rightarrow \frac{68 - 95 - 94.7}{\sigma} \rightarrow$$

↑ ↑

Rule

$$\mu = 165$$

$$\sigma = 10$$

↓↓

$$[\mu - \sigma, \mu + \sigma]$$

$$[165 - 2\sigma, 165 + 2\sigma]$$

$$[165 - 2\sigma, 185]$$

$$[145, 185]$$

↑

95.1%

$$[165 - 1\sigma, 165 + 1\sigma]$$

$$[155, 175]$$

↑

↑

68.7%

94.7%

$\uparrow x$

$\downarrow$

tag? $\rightarrow$

mean

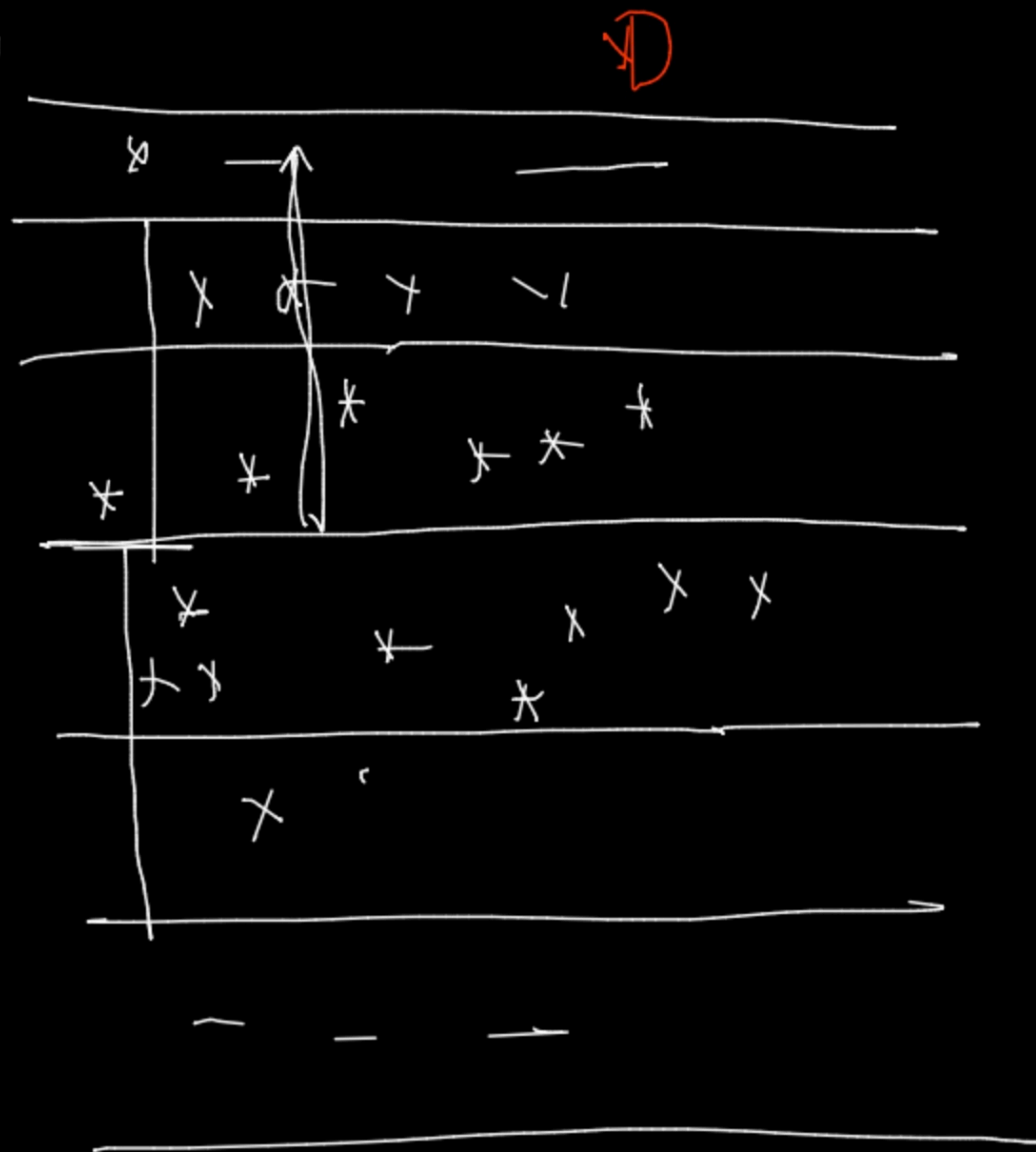
$+3\sigma$

$+2\sigma$

$-\sigma$

$+\sigma$

-3σ



$$[165 - 3\sigma, 165 + 3\sigma]$$

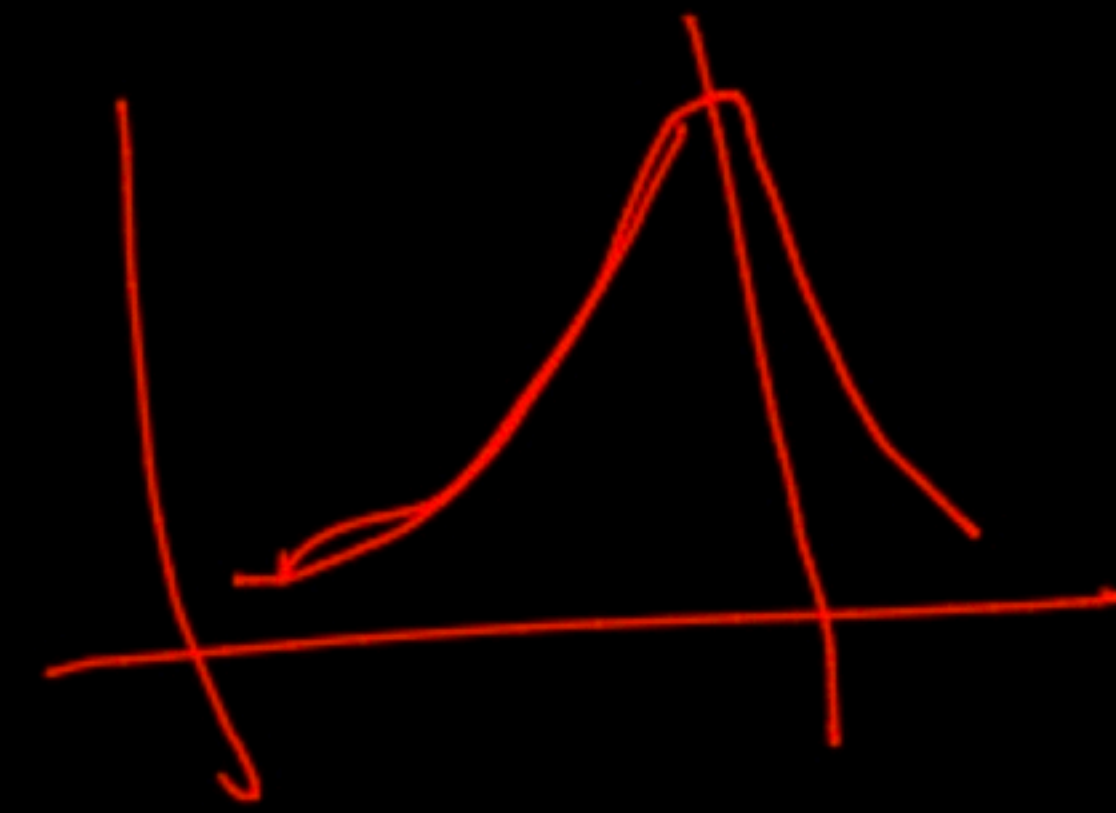
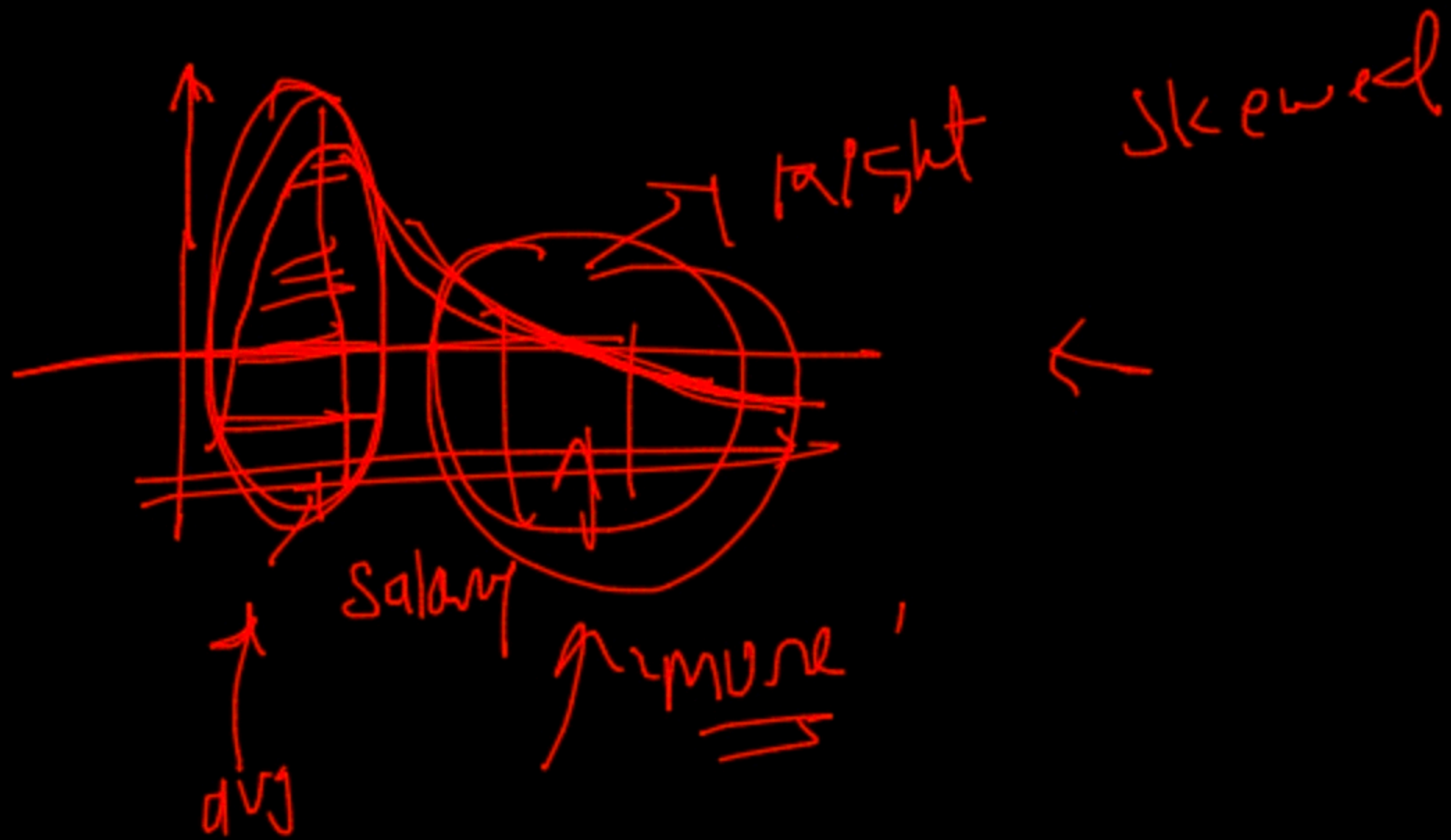
$\downarrow$

$$[135, 195]$$

$$148$$

$$[4 - 3\sigma, 4 + 3\sigma]$$

Skewness
↑



left skewed
↑

Stock
φ

ML

↑ (faster) →

Blackboard